

GIT Repository

Download and install

<https://gitforwindows.org>

We will use some simple command line code to understand version control in general.

Git & GitHub CLI Worksheet

Part 1: Setup & Initialization

Before starting, ensure you have configured your identity.

Bash

```
# Set your global identity
git config --global user.name "Your Name"
git config --global user.email "your_email@example.com"

# Create and enter your project folder
mkdir git-demo && cd git-demo

# Initialize the repository
git init -b main
```

Part 2: The Versioning Demo (Modify-Add-Commit)

We will create three versions of a single file to see how Git records history.

Version 1: The Beginning

Bash

```
# Create the file
echo "Version 1: Initial draft of my notes." > notes.txt

# Stage and commit
git add notes.txt
git commit -m "Initial commit: Added notes.txt"
```

Version 2: Adding Content

Bash

```
# Update the file
echo "Version 2: Adding more insights to notes." >> notes.txt

# Commit the change
git add notes.txt
git commit -m "Update: Added second line to notes.txt"
```

Version 3: Final Polish

Bash

```
# Final update
echo "Version 3: Concluding the thoughts." >> notes.txt

# Final commit
git add notes.txt
git commit -m "Final: Added conclusion to notes.txt"

# VIEW THE HISTORY
git log --oneline
```

Observation: You should now see three distinct snapshots (commits) in your log.

Part 3: The .gitignore Demo

This part demonstrates how to prevent "clutter" (like logs or temp files) from being tracked.

Bash

```
# 1. Create a log file that we DON'T want in Git
echo "System log data..." > system.log

# 2. Check status (Git will show system.log as 'untracked')
git status

# 3. Create a .gitignore file and tell it to ignore all .log files
echo "*.log" > .gitignore

# 4. Check status again
git status
```

Observation: Notice that `system.log` has disappeared from the "untracked files" list, but the `.gitignore` file itself is now visible and needs to be tracked.

Bash

```
# 5. Commit the .gitignore so your team shares these rules
git add .gitignore
git commit -m "Add .gitignore to exclude log files"
```

Part 4: Pushing to GitHub

To finish, we will link this local history to a remote repository.

1. **On GitHub:** Create a new repository named `git-demo`. **Do not** initialize it with a README or License.
2. **On CLI:** Copy the URL (e.g., `https://github.com/username/git-demo.git`) and run:

Bash

```
# Add the remote link
git remote add origin YOUR_GITHUB_REPO_URL
```

```
# Verify the link
git remote -v
```

```
# Push your history to GitHub
git push -u origin main
```